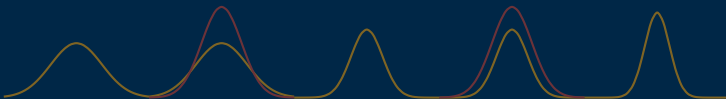


# Gated Data-Space Inversion for Reservoir Prediction

Reducing variance loss in posteriors

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Masters dissertation ▷ Thiago Tomás de Paula



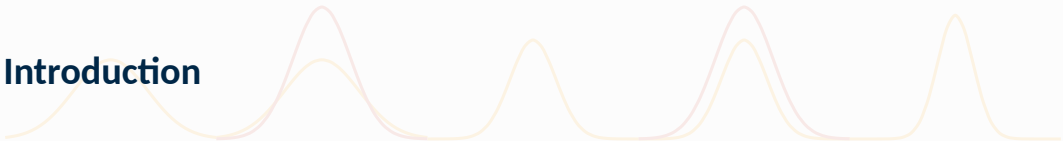
August 4th, 2026

Graduate Program in Mechanical Engineering

University of Brasília



# Introduction



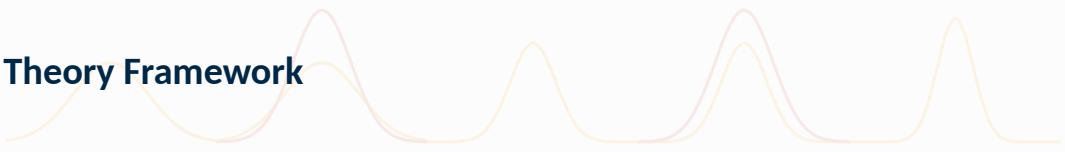








## Theory Framework

The background of the slide features a series of overlapping, wavy lines in a light yellow and a soft pink color, creating a decorative, flowing pattern across the middle of the page.



## Prediction forward model

The prediction forward model, or forward model, is the underlying mapping between a model's parameters, controls and initial states to its observable features.

Denoting by  $\mathbf{m}$  the vector of model parameters, and by  $\mathbf{u}$  the set of controls applied on the field, and by  $\mathbf{d}$  the reservoir production within the time frame of the controls, one can write

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}; \mathbf{x}_0) \quad (1)$$

with  $\mathbf{x}_0$  the initial value of reservoir states.





The key assumption behind the forward model is that it is deterministic. Implementation wise,  $g$  consists of evaluating a set of equations at each gridblock of the model.

Each evaluation evolves the model a little further in time, until the scheduled time period is appreciated, concluding the process.

Black-oil equations are the most common implementations of the forward model [2] due to their fair simplicity and accuracy. The fluid is assumed composed of three liquid phases, water, oil and gas.



A given phase  $\alpha$ 's component mass  $A_\alpha$  satisfies the conservation equation

$$\frac{\partial}{\partial t} (\phi_{\text{ref}} A_\alpha) + \nabla \cdot \mathbf{u}_\alpha + q_\alpha = 0, \quad (2)$$

where

$$A_w = m_\phi b_w s_w, \quad \mathbf{u}_w = b_w \mathbf{v}_w, \quad (3a)$$

$$A_o = m_\phi (b_o s_o + r_{og} b_g s_g), \quad \mathbf{u}_o = b_o \mathbf{v}_o + r_{og} b_g \mathbf{v}_g, \quad (3b)$$

$$A_g = m_\phi (b_g s_g + r_{go} b_o s_o), \quad \mathbf{u}_g = b_g \mathbf{v}_g + r_{go} b_o \mathbf{v}_o. \quad (3c)$$

and the phase fluxes are given by Darcy's law:

$$\mathbf{v}_\alpha = -\lambda_\alpha \mathbf{K} (\nabla p_\alpha - \rho_\alpha \mathbf{g}). \quad (4)$$

Finally, the following closure equations are imposed:

$$s_w + s_o + s_g = 1, \quad p_{c,ow} = p_o - p_w, \quad p_{c,og} = p_o - p_g. \quad (5)$$



## Predicting versus forecasting

Prediction refers to the estimation of any datum at a given time, while forecasting refers to the estimation of any datum beyond the observation period. Forecasting is a type of prediction, but in particular prediction of new data.

## History and validation

In Data Science literature, a common way of estimating forecast quality is splitting available data into a training partition and a test partition. The training partition is used to freely adjust predictions within a given period, and the test, to verify how close predictions will be on new data.

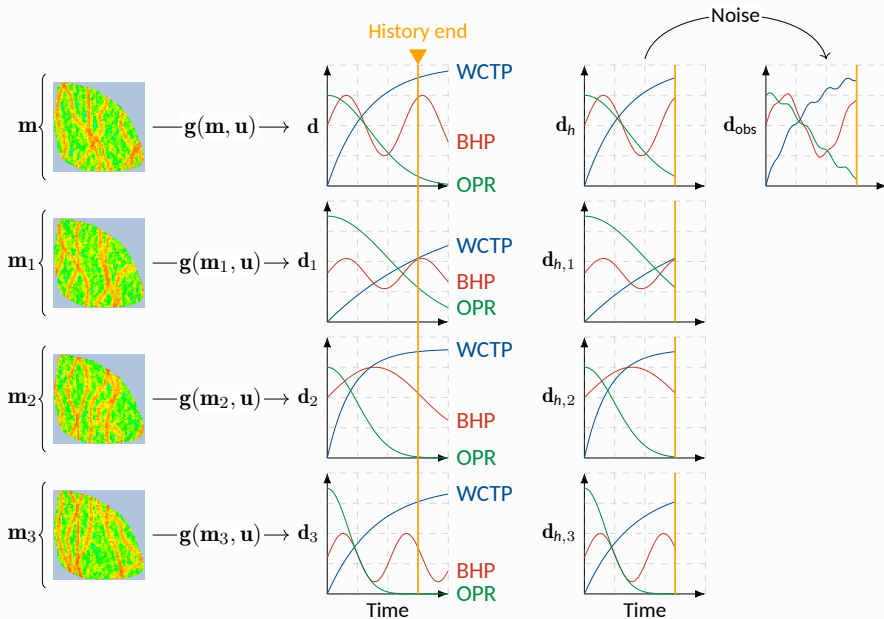
In the context of this work, data from a given reference will be split into two disjointed periods in time: history period, and validation period, mirroring the common train and test splits respectively.



Let  $\mathbf{d}$  denote the true production data vector of a reference reservoir across time, free of observation error. Data is separated into a history period and a validation period, so that  $\mathbf{d} = [\mathbf{d}_h^\top \quad \mathbf{d}_v^\top]^\top$  with  $\mathbf{d}_h$  the ground-truth history data, and  $\mathbf{d}_v$  the ground-truth validation data.

Analogously, let  $\mathbf{u}$  and  $\mathbf{m}$  be the controls applied to the ground-truth and the parameters of the ground-truth, respectively. Denoting the prediction forward model by  $\mathbf{g}(\cdot, \cdot)$ , ground-truth data is given by  $\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u})$ .

Assume  $\mathbf{d}_h$  is corrupted by additive measurement noise  $\mathbf{e}_d \sim N(\mathbf{0}, \mathbf{C}_e)$ , where  $\mathbf{C}_e$  is the noise covariance matrix. The corrupted vector is the observed data during the history period, denoted  $\mathbf{d}_{\text{obs}}$ .







$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \quad (6a)$$

$$\mathbf{K}^{\ell} = \Delta \mathbf{D}^{\ell} \Delta \mathbf{D}_h^{\ell} (\Delta \mathbf{D}_h^{\ell} \Delta \mathbf{D}_h^{\ell} + \alpha^{\ell} \mathbf{C}_e)^{-1}. \quad (6b)$$



$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \quad (7)$$

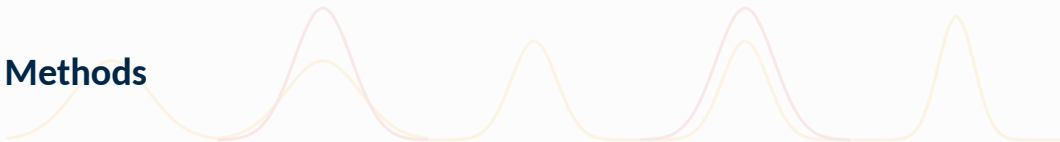


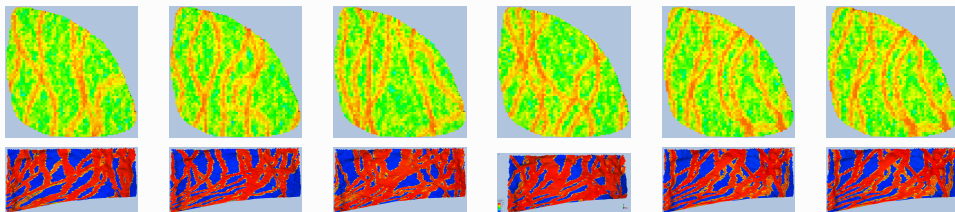


$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^\ell + (\mathbf{R} \circ \mathbf{K}^\ell)[\boldsymbol{\alpha}_j^\ell \circ (\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^\ell)], \quad (8a)$$

$$\alpha_{j,v}^\ell = \text{erf} \left( \frac{1}{\lambda \sqrt{2\sigma_v^2}} |(\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^\ell)_v| \right) \quad (8b)$$

## Methods

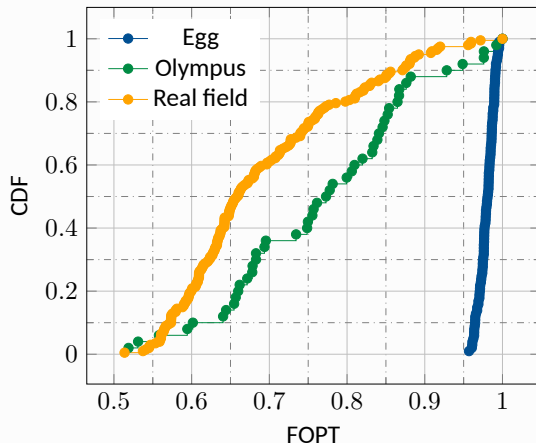


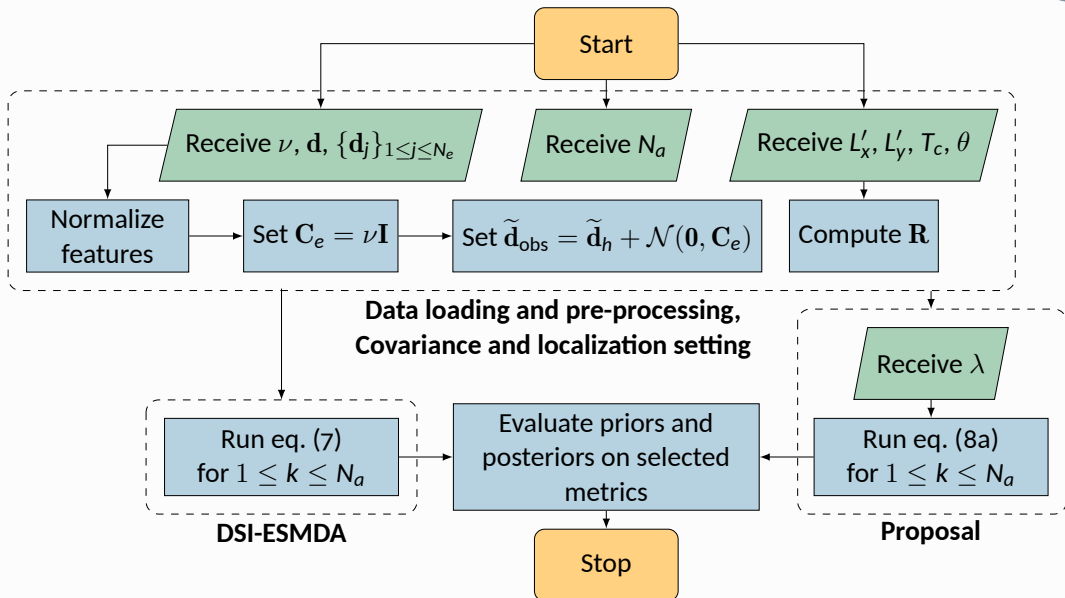


| Property                | Egg                   | Olympus                  | Real field | Unit  |
|-------------------------|-----------------------|--------------------------|------------|-------|
| Rock compressibility    | 0.0                   | $1.42 \times 10^{-10}$   | -          | 1/Pa  |
| Water compressibility   | $1.0 \times 10^{-10}$ | $3.97 \times 10^{-10}$   | -          | 1/Pa  |
| Oil dynamic viscosity   | $5.0 \times 10^{-3}$  | $2.8-3.5 \times 10^{-3}$ | -          | Pa·s  |
| Water dynamic viscosity | $1.0 \times 10^{-3}$  | $0.398 \times 10^{-3}$   | -          | Pa·s  |
| Well diameter           | 20                    | 19                       | -          | cm    |
| Wells                   | 8(P) 4(I)             | 11(P) 7(I)               | 16(P) 8(I) | -     |
| Active cells            | 18,533                | 192,750                  | -          | -     |
| Cell size               | $8 \times 8 \times 4$ | $50 \times 50 \times 3$  | -          | m×m×m |
| Ensemble size           | 101                   | 50                       | 201        | -     |



The field oil production total (FOPT) is a measurement commonly used to compare ensembles since ground-truth and realizations are each represented by a single value: how much oil was produced within the period of interest.







Four metrics are considered on the evaluation of posterior/prior distributions [1, subsection 9.4.2]: history observation coverage, validation observation coverage, mean squared error, and data-mismatch.

Denoting an ensemble by  $\mathbf{Y} = [\mathbf{y}_1 \ \cdots \ \mathbf{y}_{N_e}]$ , these metrics are defined as:

$$\text{OC}_h(\mathbf{Y}) = \frac{1}{N_h} \sum_{i=1}^{N_h} \mathbb{1} (\min(\mathbf{Y})_i \leq \mathbf{d}_{h,i} \leq \max(\mathbf{Y})_i) \quad (9)$$

$$\text{OC}_v(\mathbf{Y}) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{1} (\min(\mathbf{Y})_i \leq \mathbf{d}_{v,i} \leq \max(\mathbf{Y})_i) \quad (10)$$

$$\text{MSE}(\mathbf{Y}) = \frac{1}{N_h} (\mathbf{d}_h - \bar{\mathbf{y}})^\top \mathbf{C}^{-1} (\mathbf{d}_h - \bar{\mathbf{y}}), \quad \mathbf{C} = \Delta \mathbf{Y} \Delta \mathbf{Y}^\top + \mathbf{C}_e \quad (11)$$

$$\mathcal{O}(\mathbf{y}_j) = \frac{1}{2N_h} (\mathbf{d}_h - \mathbf{y}_{j,h})^\top \mathbf{C}_e^{-1} (\mathbf{d}_h - \mathbf{y}_{j,h}) \quad (12)$$

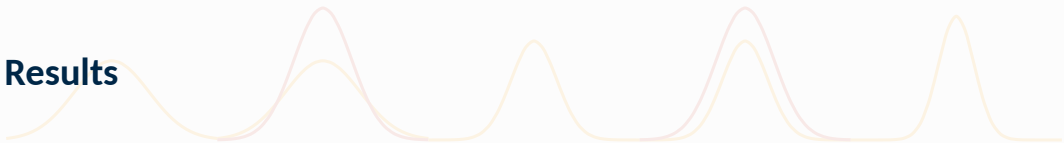


| Reservoir  | $N_e$ | Number of time steps | Time step size   | Simulation life       | History split |
|------------|-------|----------------------|------------------|-----------------------|---------------|
| Egg        | 100   | 120                  | 30 days          | 3600 days (9.8 years) | 60%           |
| Olympus    | 49    | 20                   | 1 calendar year  | 7305 days (20 years)  | 60%           |
| Real field | 200   | 510                  | 1 calendar month | 12785 days (35 years) | 60%*          |

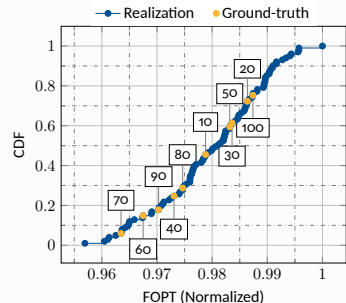
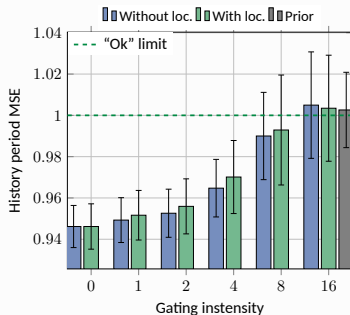
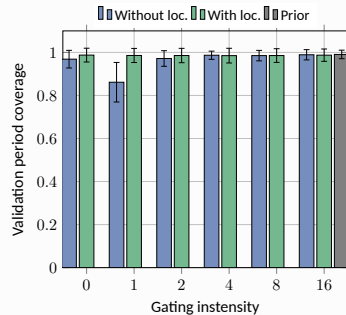
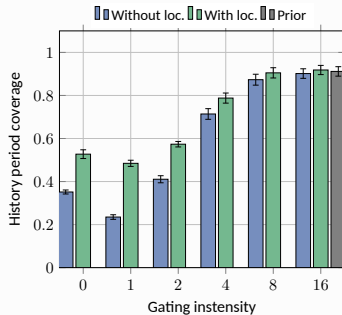
| Reservoir  | Critical length in x | Critical length in y | Critical time | Anisotropy |
|------------|----------------------|----------------------|---------------|------------|
| Egg        | 100                  | 100                  | 3000 days     | 0°         |
| Olympus    | 100                  | 100                  | 7200 days     | 0°         |
| Real field | -                    | -                    | 10500 days    | 0°         |

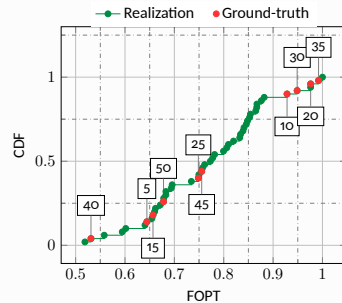
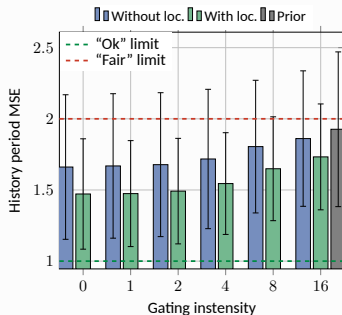
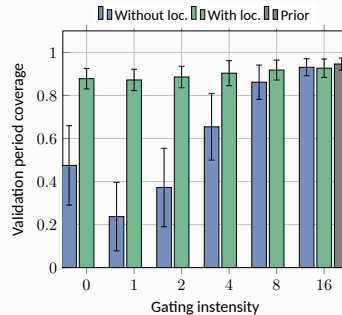
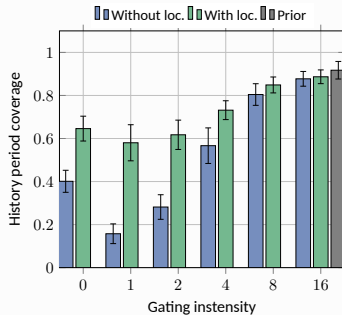
| Reservoir  | Well features used<br>Producer | Well features used<br>Injector | Number of wells<br>Producer | Number of wells<br>Injector | Total features           |
|------------|--------------------------------|--------------------------------|-----------------------------|-----------------------------|--------------------------|
| Egg        | OPR, WPR                       | BHP                            | 4                           | 8                           | 16                       |
| Olympus    | OPR, WPR                       | WIR                            | 11                          | 7                           | 29                       |
| Real field | BHP, WCTP                      | BHP                            | 16                          | 8                           | 39 (a WCTP was constant) |

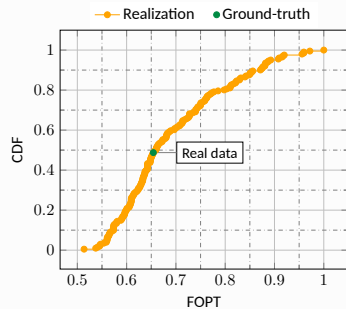
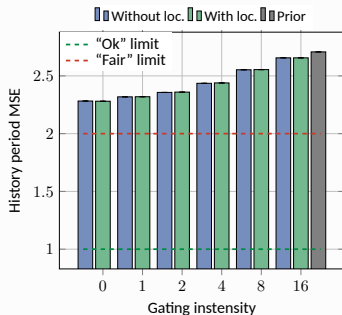
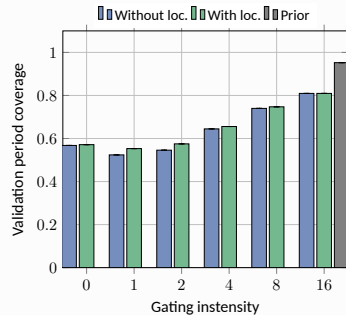
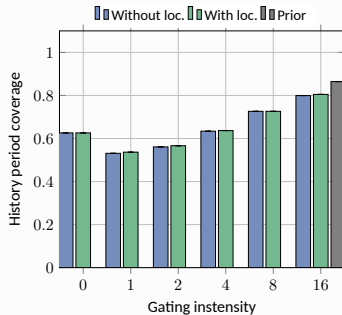
## Results



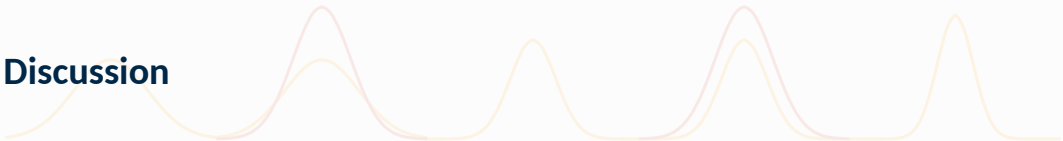








## Discussion





### OC, MSE and mismatch increase

Increase in gating intensity tended to increase observation coverage, mean squared error, and data mismatch, albeit with case-dependent sensitivities.

### OC, MSE and mismatch increase

Localization yielded mixed results: for the Egg and real field cases, effects on coverage and MSE were minimal, especially for intense enough gatings, while the effect in the Olympus case was drastic. Cumulative distribution plots of the field oil production total suggest that ground-truths are unbiased in the Egg and real field cases, but somewhat biased for the Olympus case.

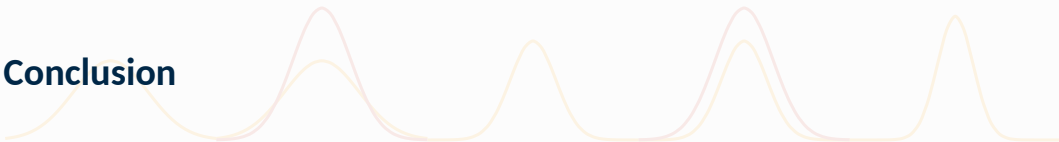
### Mild and small gating intensities versus standard DSI-ESMDA

Mild gating intensities (2 in the Egg and Olympus cases, 4 in the real field case) produced similar results to the standard DSI-ESMDA for same localization configuration, while small gating (1 for the Egg case, 1 and 2 for the Olympus and real field cases) had worse coverages than the standard DSI-ESMDA.





**Conclusion**















- [1] Alexandre Anozé Emerick. *Ensemble Data Assimilation Applied to Geological Reservoir Models*. 1st ed. Rio de Janeiro, RJ, Brazil: Universidade Petrobras, June 2025. ISBN: 978-65-88763-29-2.
- [2] Atgeirr Flø Rasmussen et al. "The Open Porous Media Flow reservoir simulator". In: *Computers & Mathematics with Applications* 81 (2021). Development and Application of Open-source Software for Problems with Numerical PDEs, pp. 159–185. ISSN: 0898-1221. DOI: <https://doi.org/10.1016/j.camwa.2020.05.014>.

# Gated DSI-ESMDA

Reducing variance loss in posteriors

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August 4th, 2026

*“It’s hard to make predictions, specially about the future”*

— Yogi Berra

**Thank you!**